INDIRA COLLEGE OF COMMERCE AND SCIENCE

Placement Salary Prediction and Analysis Using Machine Learning

S.Y M.Sc. (Comp.Sci) - Sem IV Academic Year 2025-26

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1. Introduction / Background and Context

Campus placement is a critical milestone for students completing their higher education. This project analyzes a comprehensive dataset of 1,000 student records containing 26 features including academic performance, technical skills, professional experience, and soft skills. The analysis employs both classification models to predict placement status and regression models to predict salary packages for placed candidates.

2. Problem Statement

This project aims to develop predictive models that can accurately classify students as "Placed" or "Not Placed" and further predict the expected salary for placed candidates based on academic performance, technical skills, work experience, and other factors.

3. Objectives of the Study

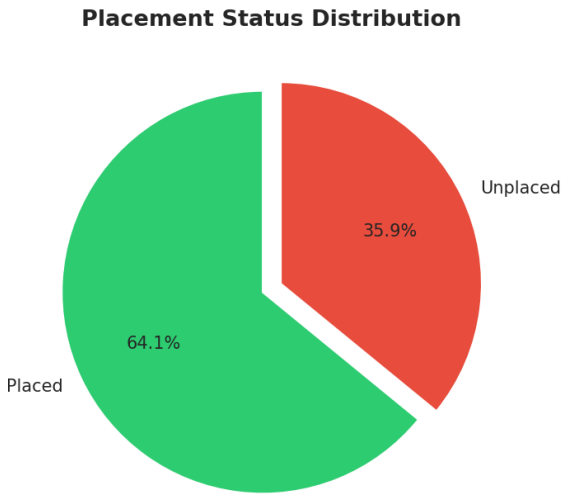
- To perform comprehensive data preprocessing on campus placement records.
- To conduct Exploratory Data Analysis (EDA) to uncover patterns and relationships.
- To implement and compare multiple ML models for placement status classification.
- To build regression models for salary prediction.
- To identify the most influential features that drive placement success.

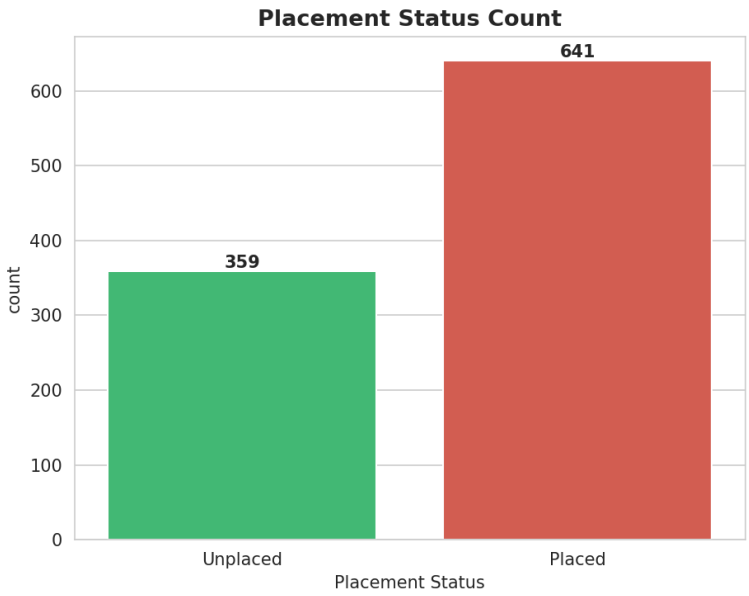
4. Dataset Description

The dataset contains 1000 records with 26 features. Placed: 641, Unplaced: 359. Salary range: 2.56 - 25.00 LPA (Mean: 14.25 LPA).

6. Exploratory Data Analysis

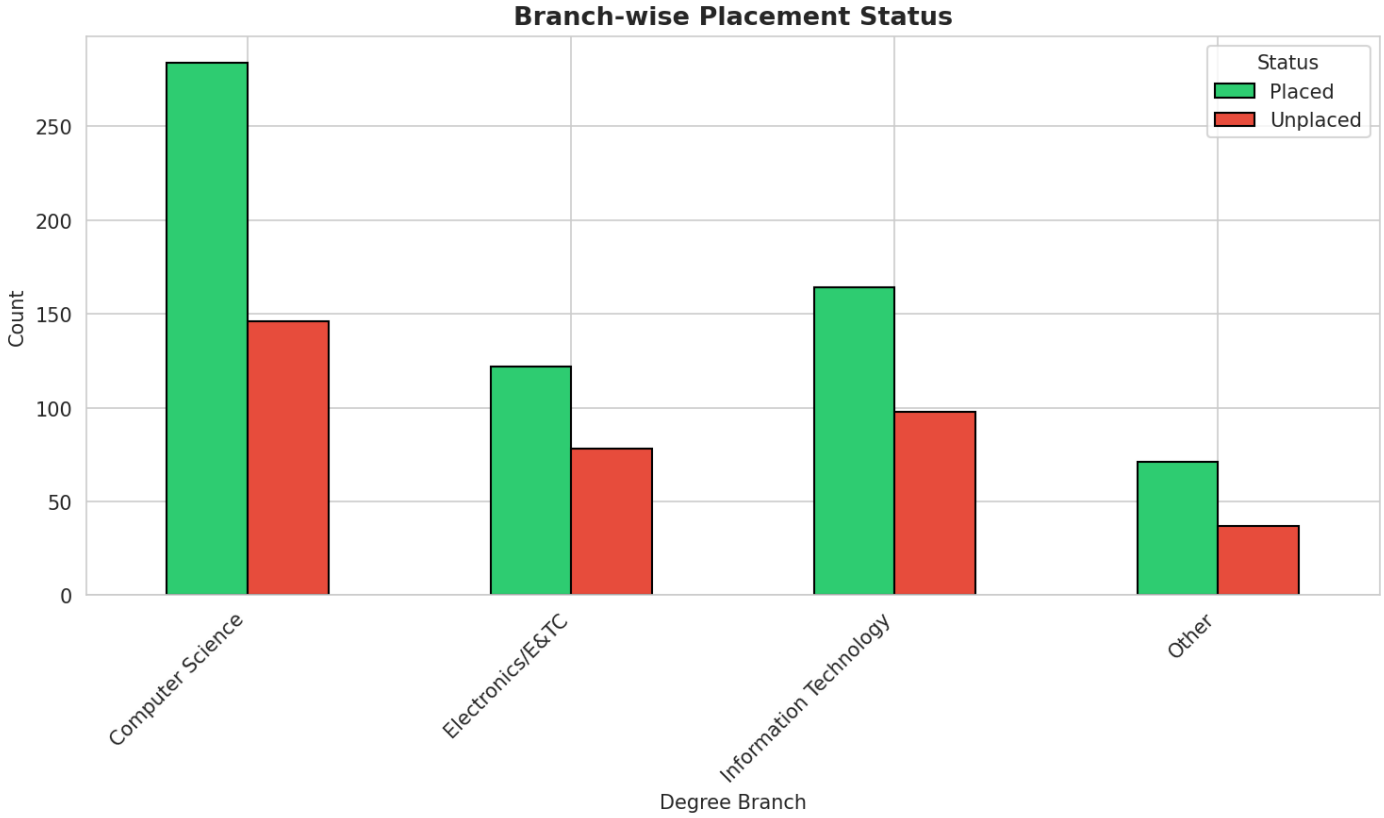
Placement status distribution.





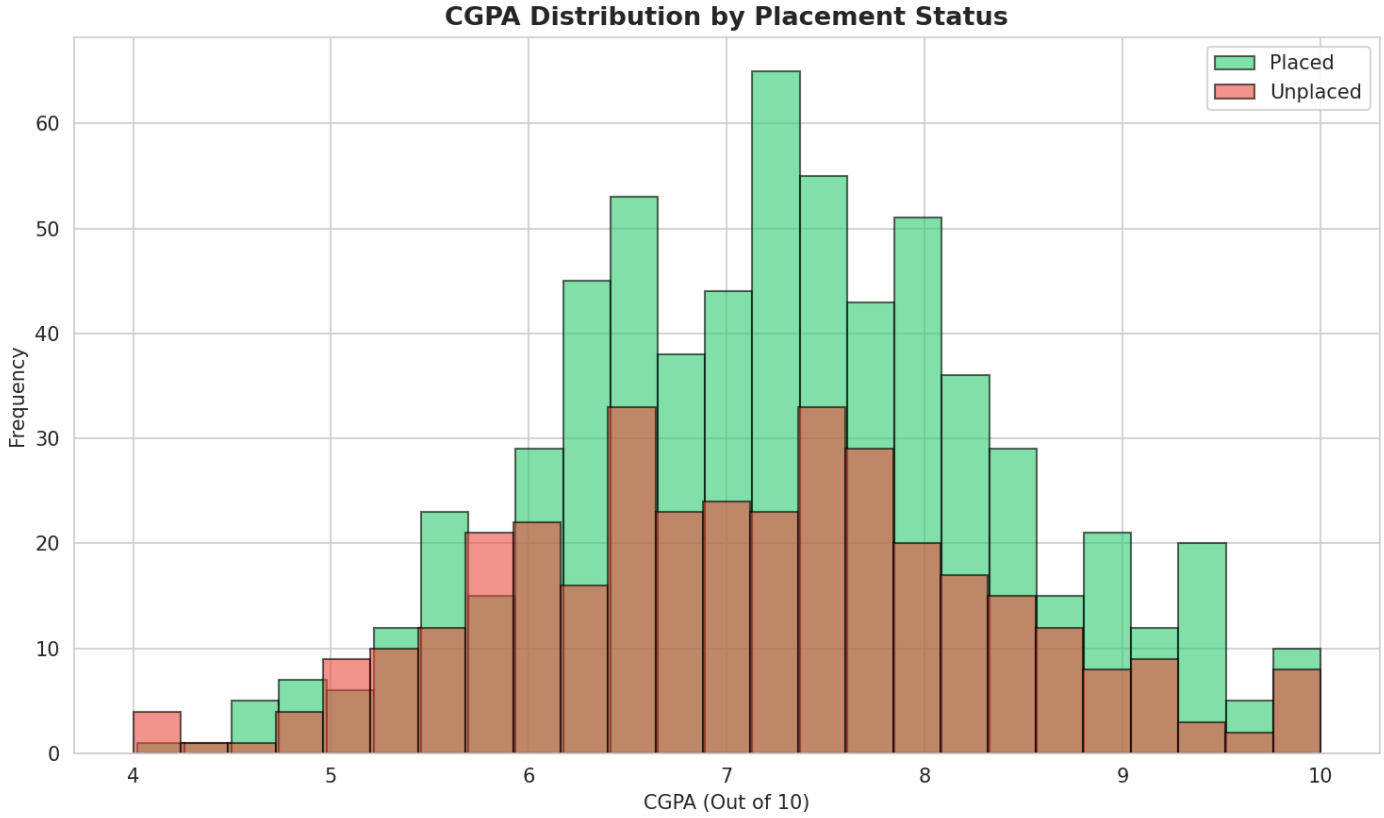
6.1 Branch-wise Placement

Distribution across degree branches.



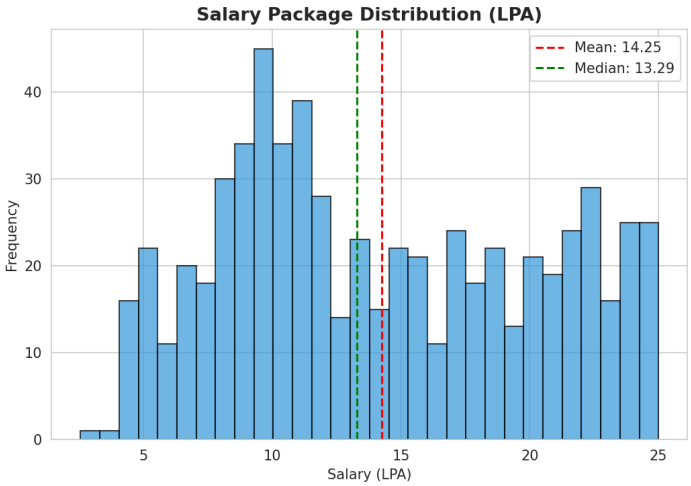
6.2 CGPA Distribution

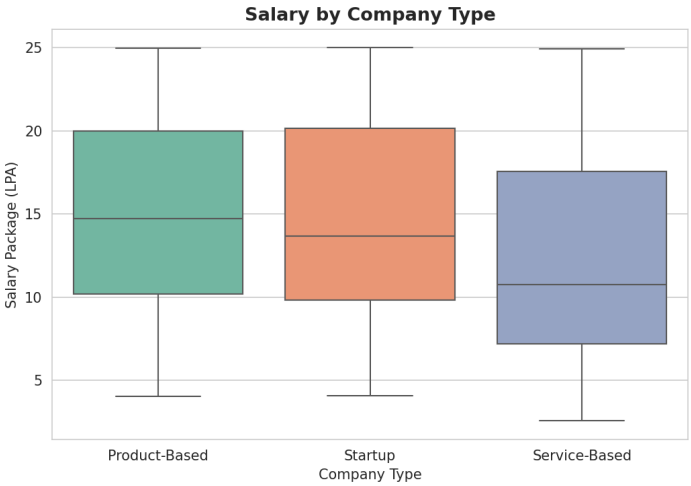
CGPA distribution by placement status.



6.3 Salary Distribution

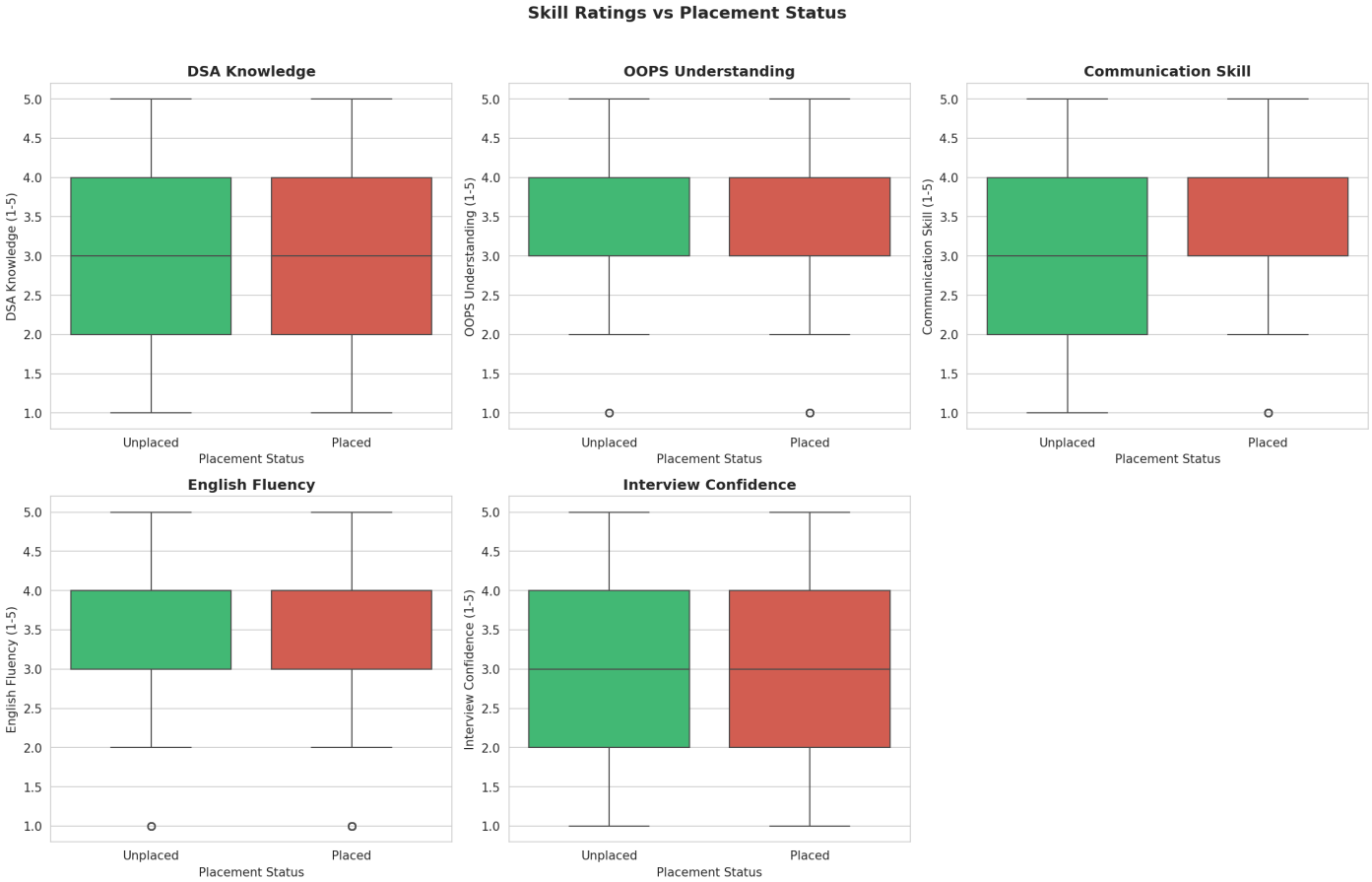
Salary package distribution and by company type.





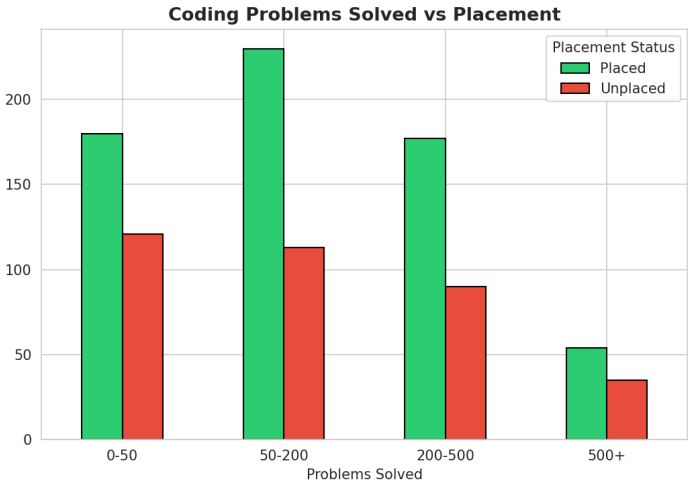
6.4 Skills vs Placement

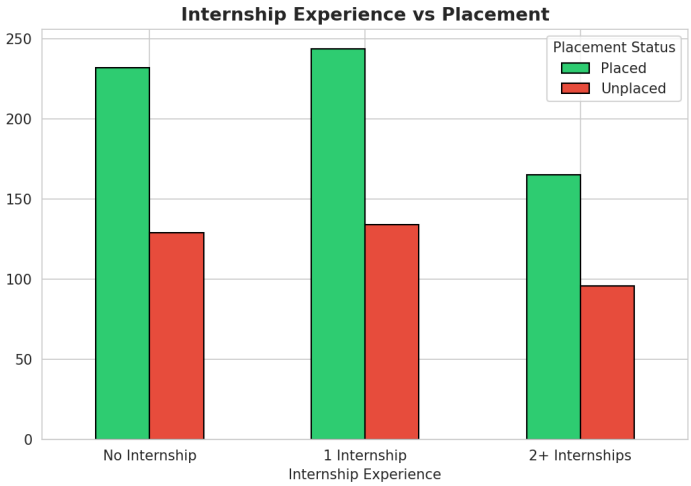
Skill ratings comparison.



6.5 Coding & Internship

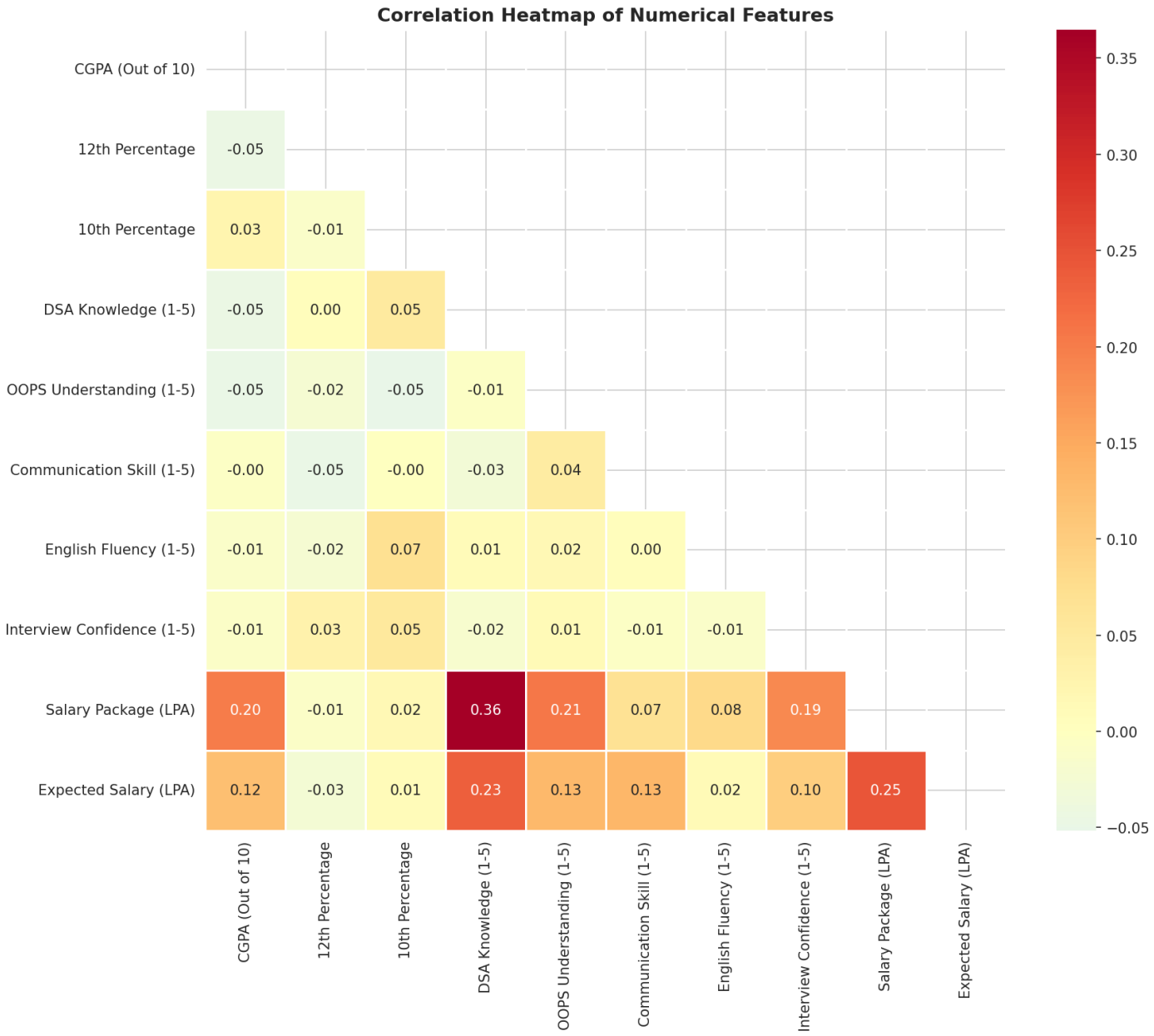
Impact on placement.





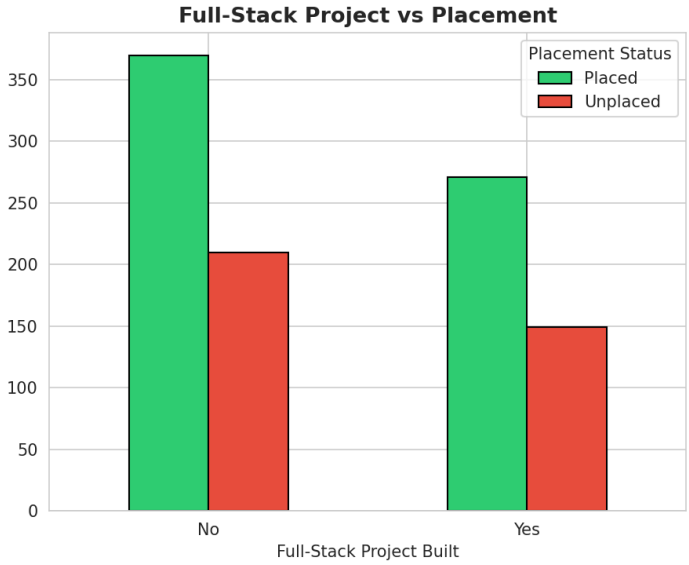
6.6 Correlation Heatmap

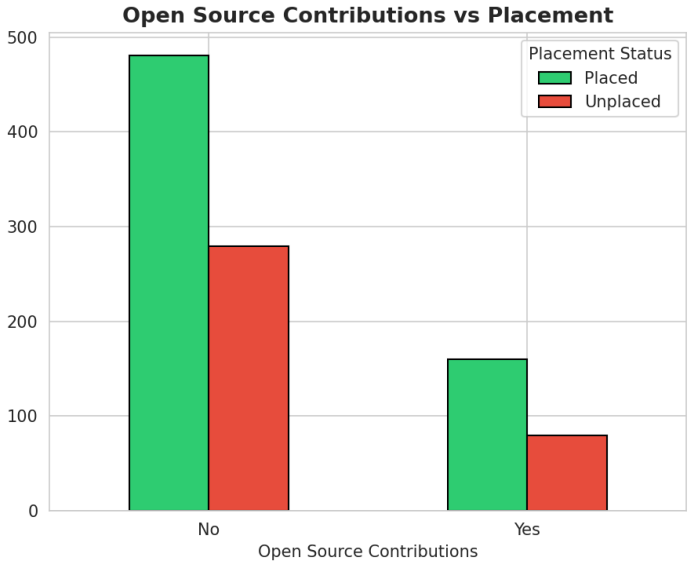
Feature correlations.



6.7 Projects & Open Source

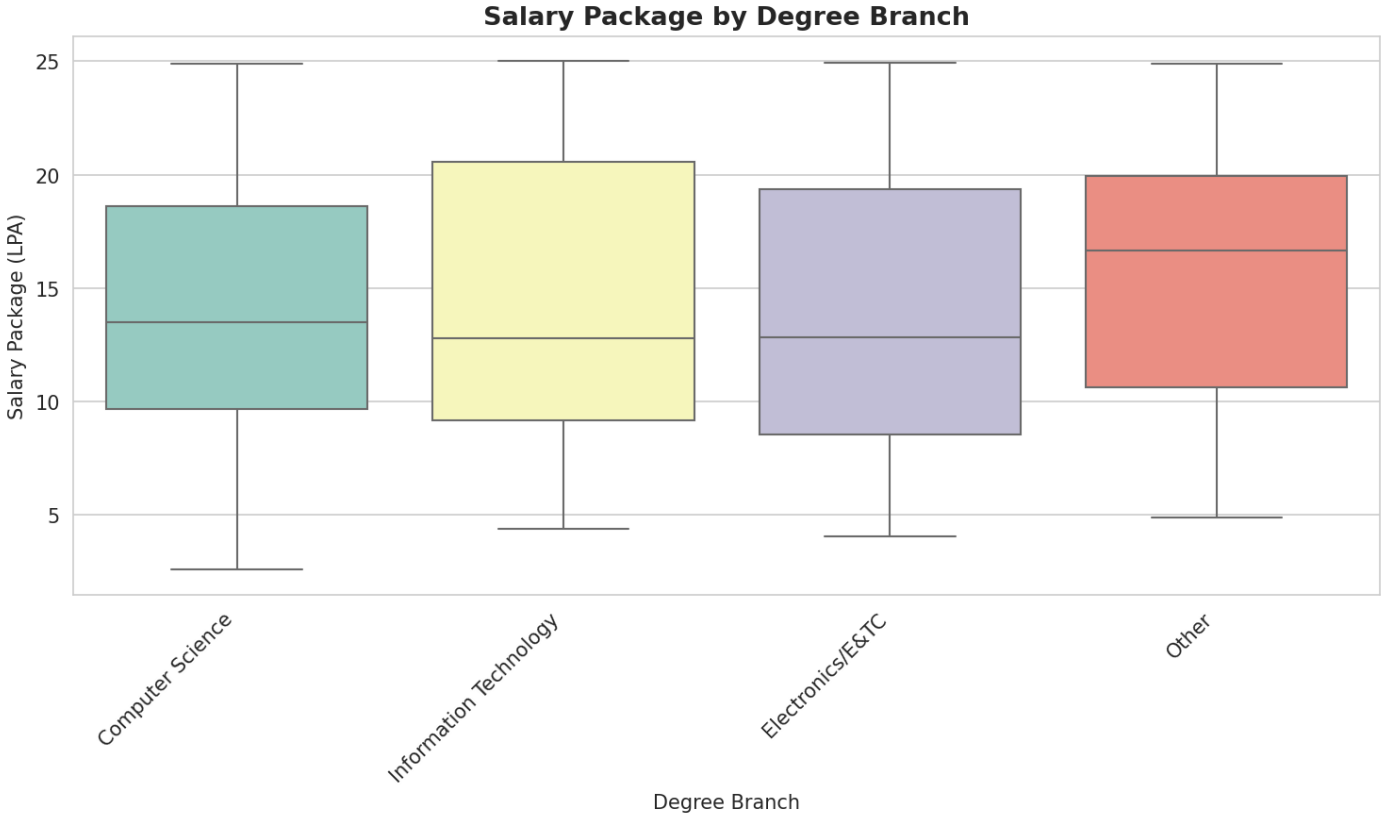
Impact on placement.





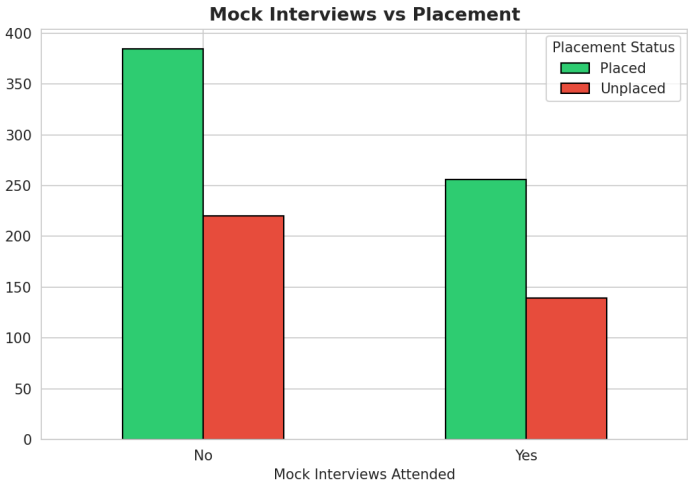
6.8 Salary by Branch

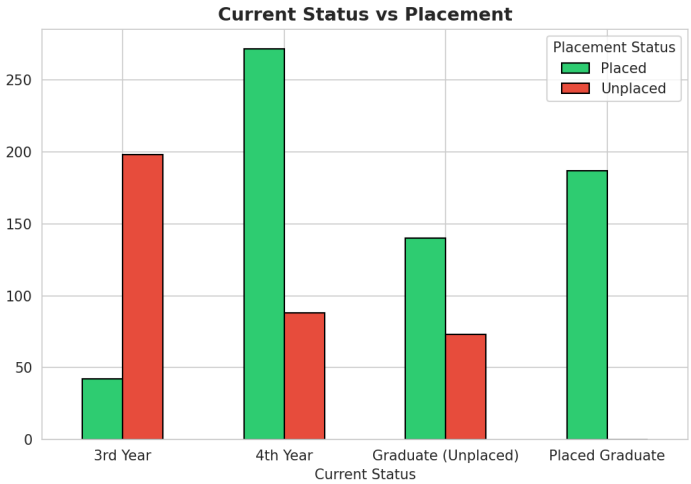
Salary comparison across branches.



6.9 Mock Interviews & Status

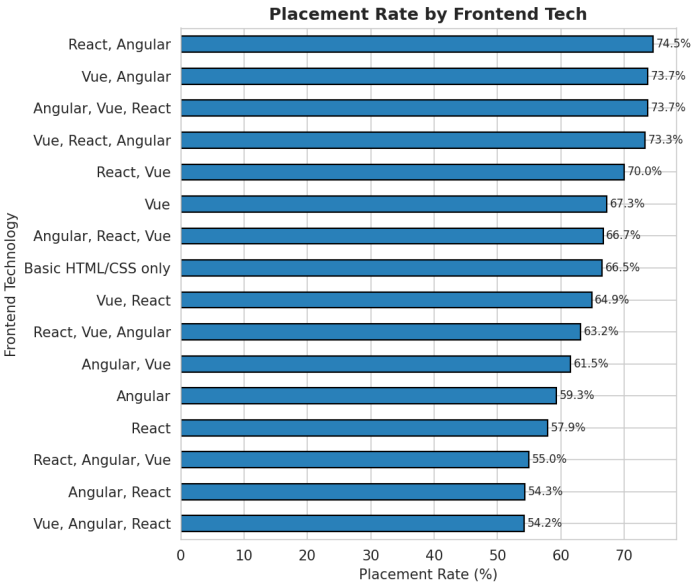
Impact on placement.

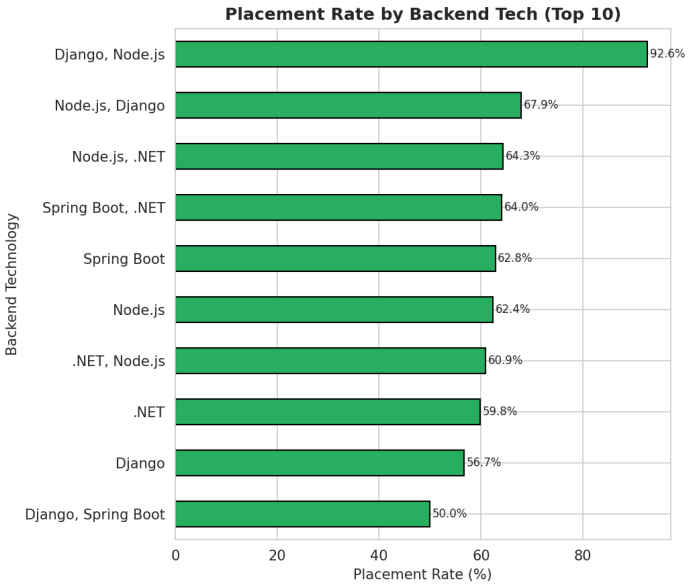




6.10 Tech Stack Analysis

Placement rates by technology.





7. Classification Results

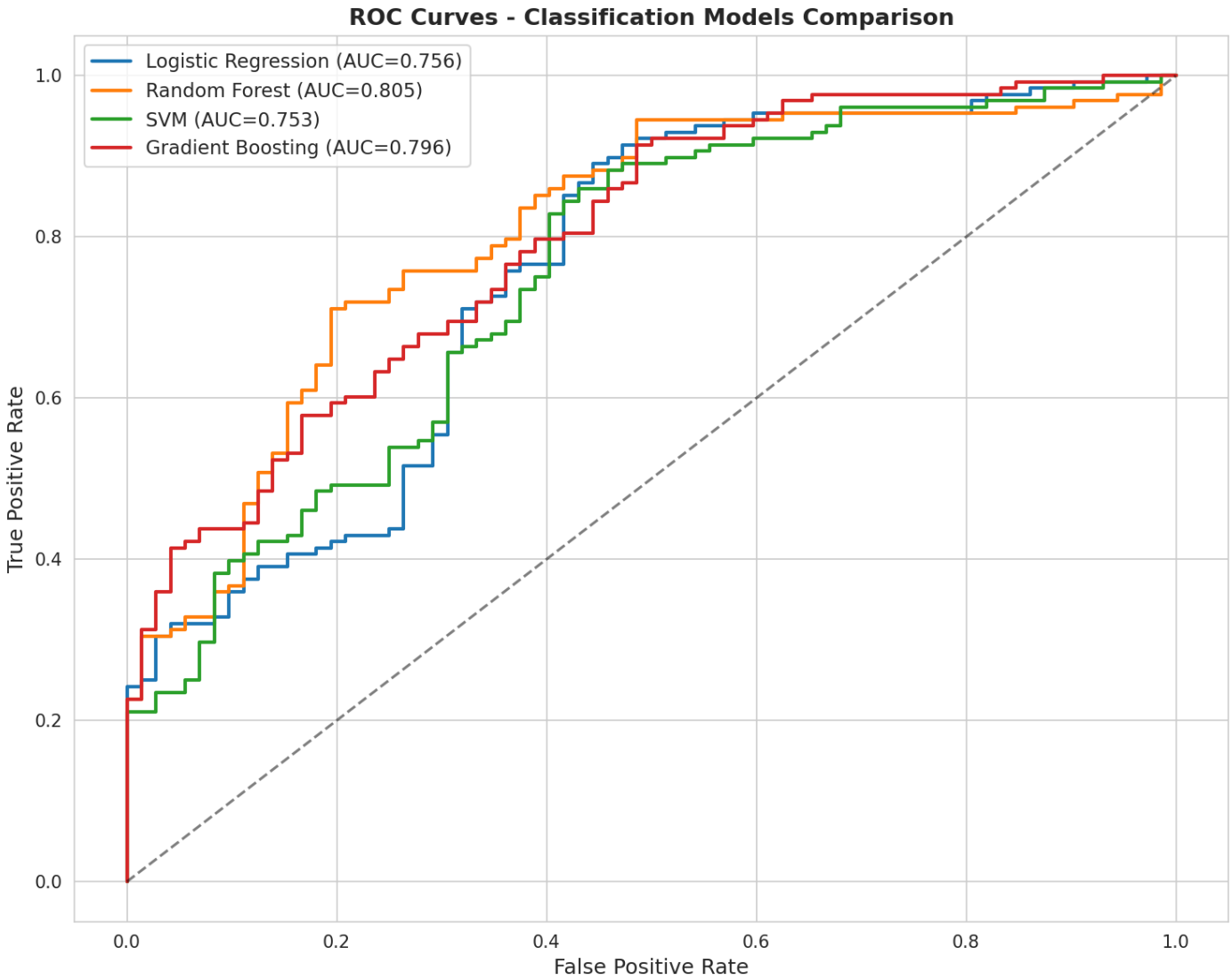
Four classification models were trained and evaluated:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	CV Accuracy
Logistic Regression	0.7700	0.7770	0.8984	0.8333	0.7560	0.7675
Random Forest	0.7750	0.7712	0.9219	0.8399	0.8052	0.7987
SVM	0.7500	0.7708	0.8672	0.8162	0.7527	0.7425
Gradient Boosting	0.7400	0.7603	0.8672	0.8102	0.7961	0.7500

Best Model: Random Forest with 77.50% accuracy.

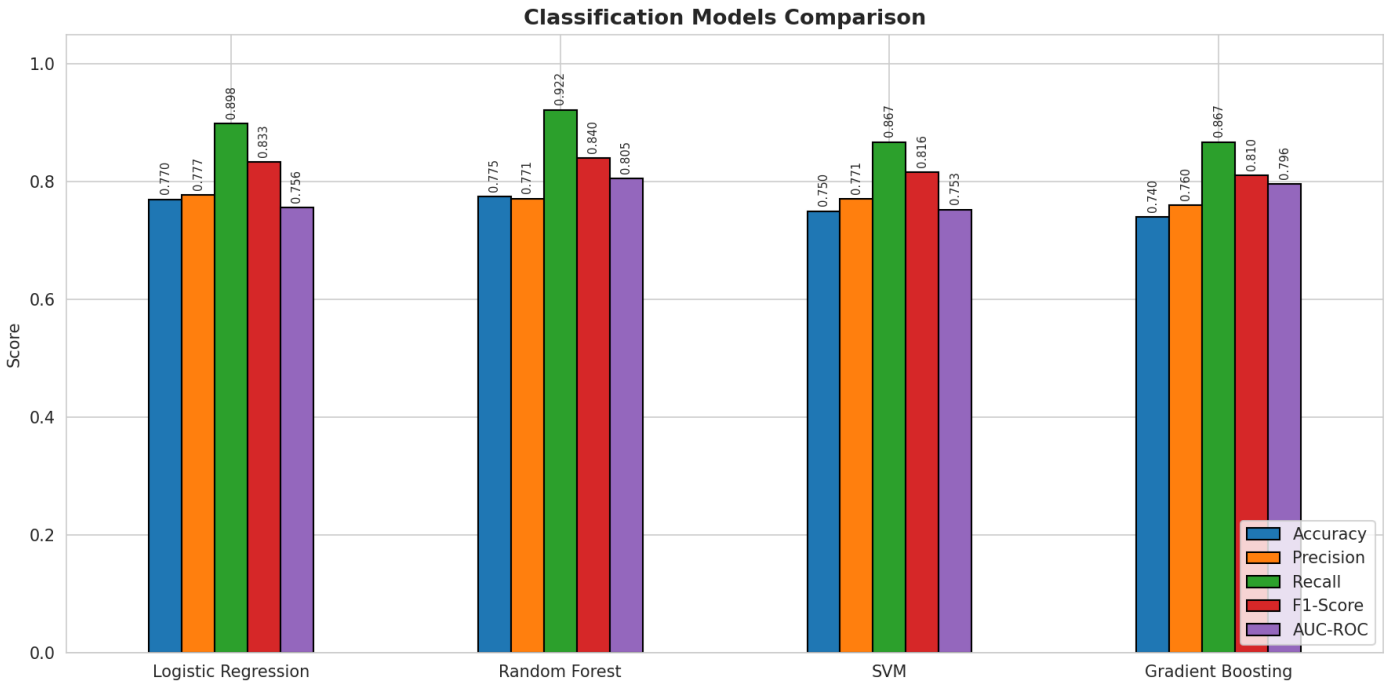
7.1 ROC Curves

ROC curves comparison.



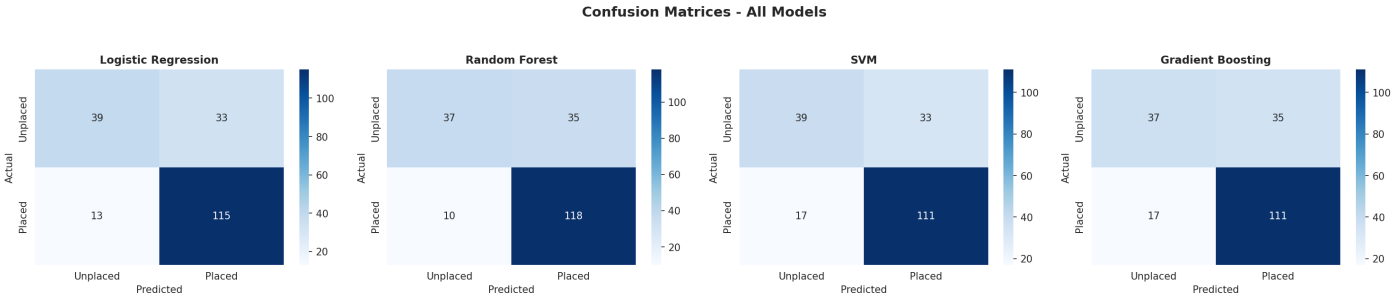
7.2 Metrics Comparison

All metrics compared.



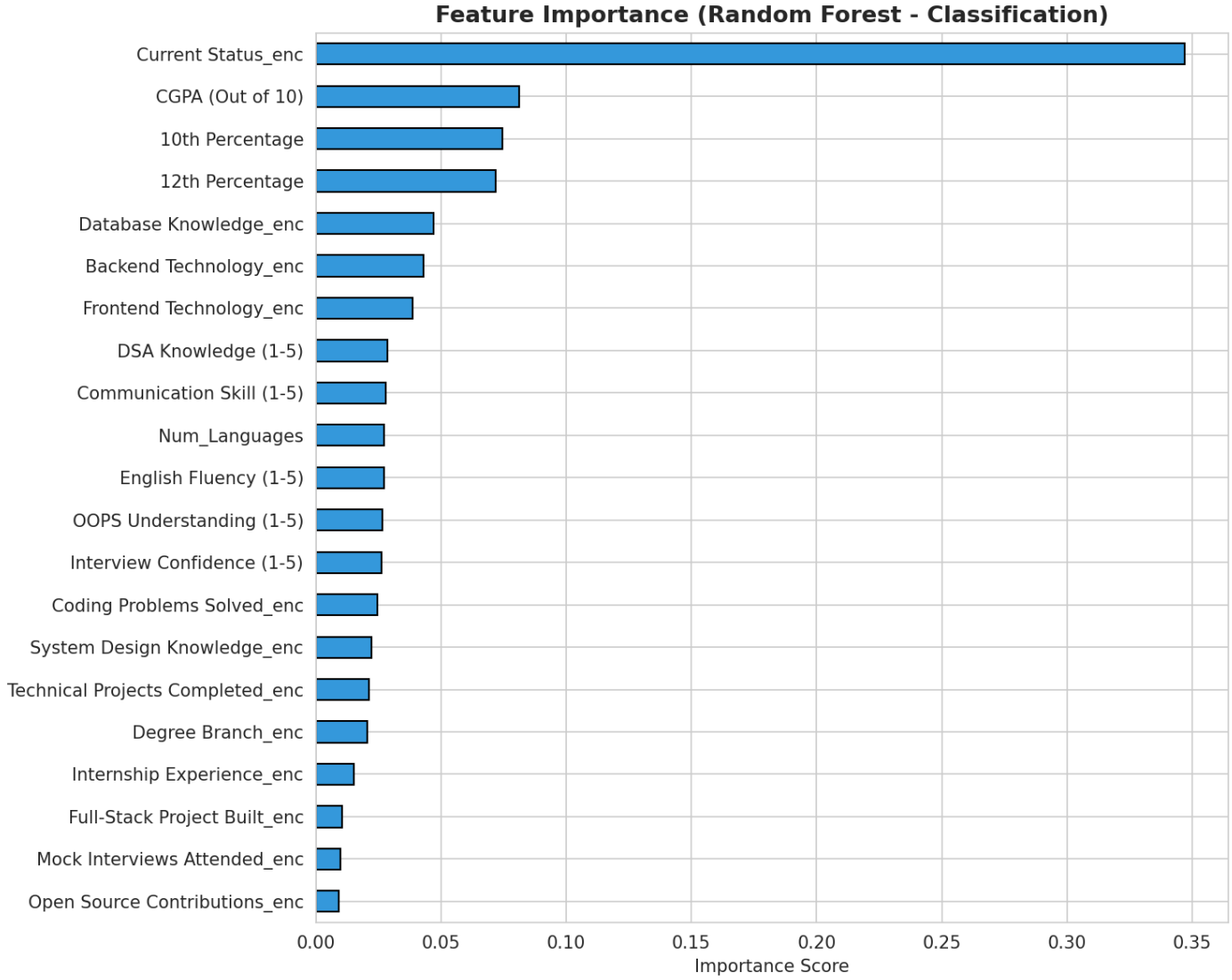
7.3 Confusion Matrices

Per-model confusion matrices.



7.4 Feature Importance

Top features: Current Status_enc, CGPA (Out of 10), 10th Percentage.



8. Regression Results

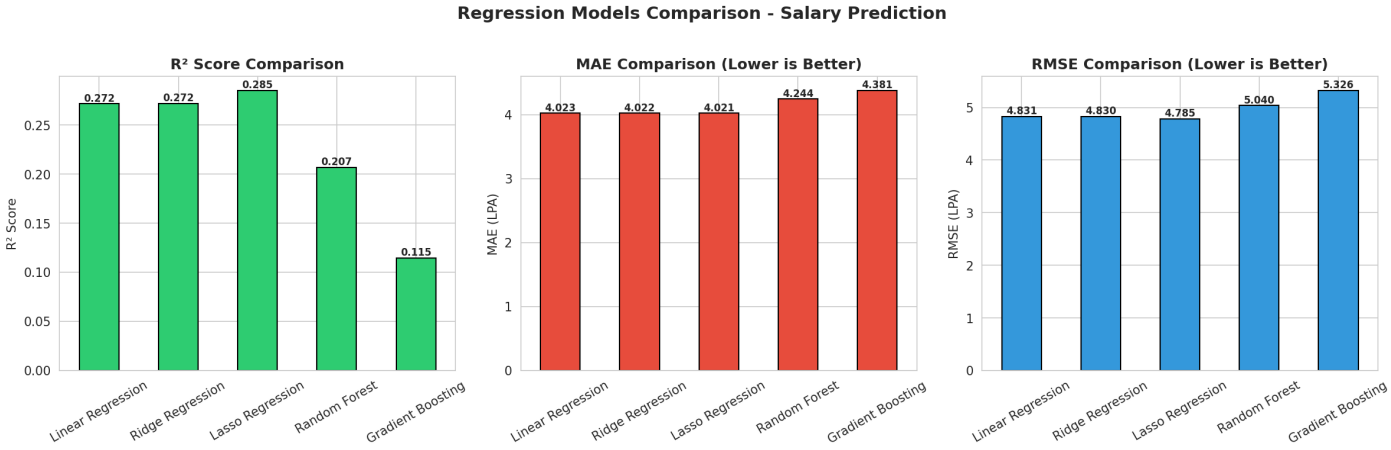
Five regression models were trained for salary prediction:

Model	R-squared	MAE (LPA)	RMSE (LPA)	CV R-squared
Linear Regression	0.2715	4.0227	4.8308	0.2418
Ridge Regression	0.2718	4.0220	4.8297	0.2420
Lasso Regression	0.2852	4.0211	4.7850	0.2482
Random Forest	0.2069	4.2436	5.0405	0.1618
Gradient Boosting	0.1145	4.3806	5.3259	0.0740

Best Model: Lasso Regression (R^2=0.2852, MAE=4.0211 LPA).

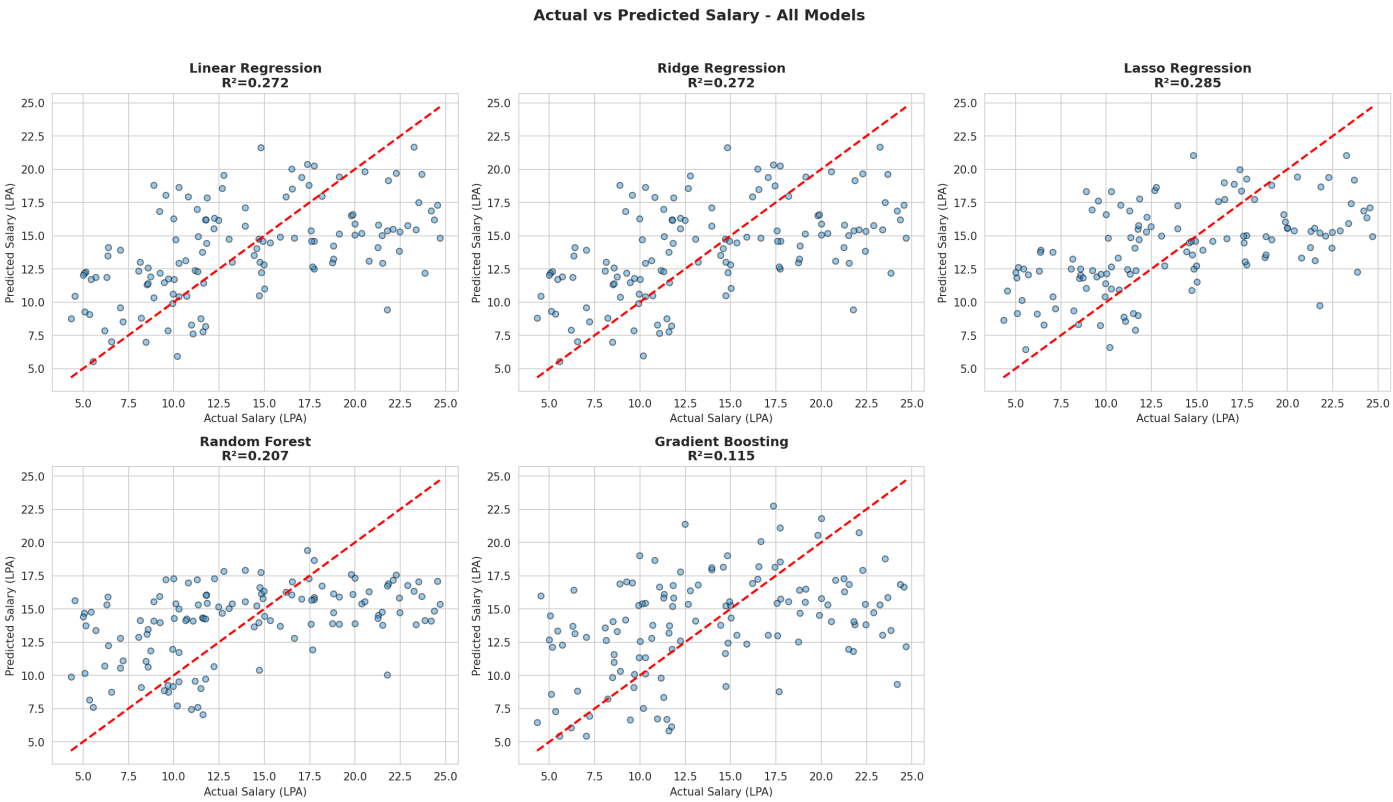
8.1 Regression Comparison

R2, MAE, RMSE compared.



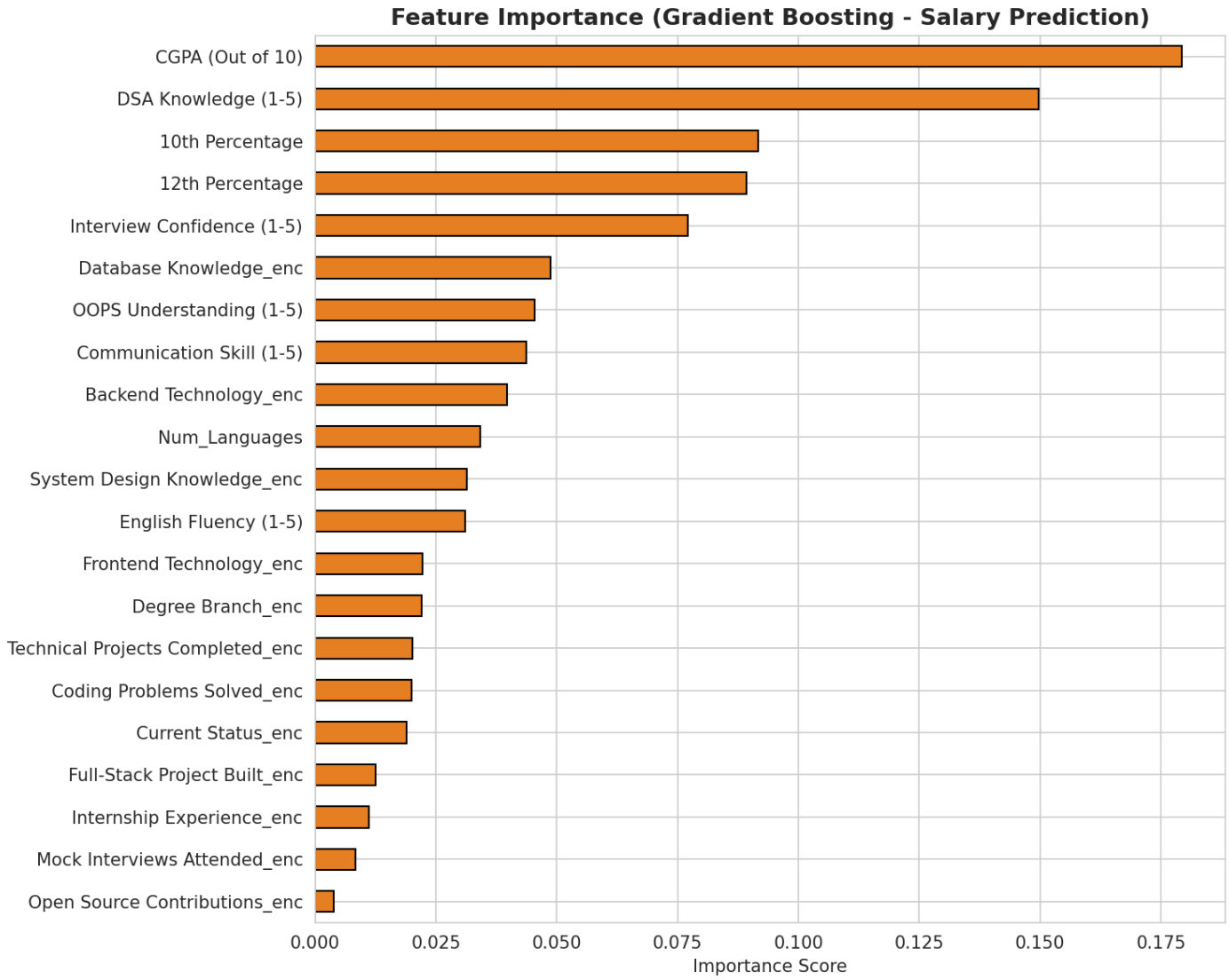
8.2 Actual vs Predicted

Scatter plots per model.



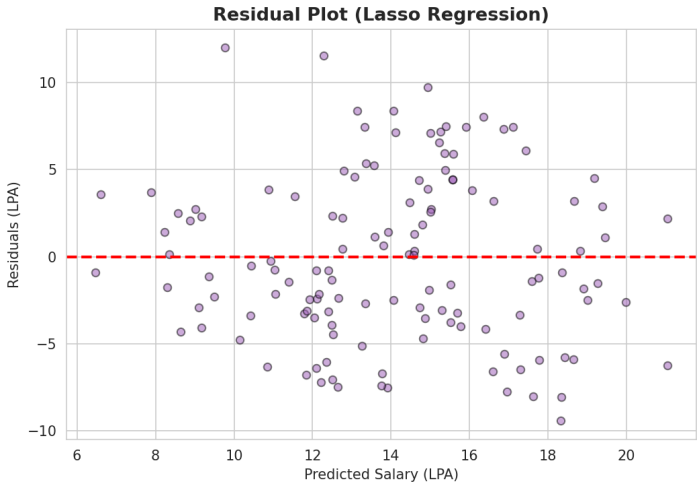
8.3 Feature Importance

Top features: CGPA (Out of 10), DSA Knowledge (1-5), 10th Percentage.



8.4 Residual Analysis

Best model: Lasso Regression.



8.5 Salary Gap

Expected vs actual salary gap.

9. Key Findings and Conclusions

- Placed students have higher average CGPA (7.29) compared to unplaced (7.10).
- Students with 2+ internships show significantly higher placement rates.
- Solving 500+ coding problems is strongly correlated with placement success.
- Product-based companies offer the highest salary packages.
- The average gap between expected and actual salary is 1.04 LPA.

10. Recommendations

- Focus on improving DSA skills, solving coding problems (target 500+), and gaining internship experience.
- Full-stack project experience and knowledge of modern technologies significantly improve placement prospects.
- Institutions should implement early identification systems for at-risk students.
- Mock interview programs should be expanded as they correlate with improved outcomes.
- Students should calibrate salary expectations based on data-driven insights.